

Description of files in this folder

1. run_trim_galore.sh
 - a. Command used to run TrimGalore on the samples
 - b. FASTQs are available under NCBI BioProject ID PRJNA1328687
2. run_kallisto.sh
 - a. Command used to run kallisto after trimming
3. sample_info_mouse_brain_rnaseq.xlsx
 - a. Metadata used for the analyses below
4. pca_and_deseq2.Rmd
 - a. Inputs:
 - i. The Kallisto output files which are available on GEO (GSE341280)
 - ii. The t2g.txt that came with the standard mouse transcriptome index that was used by kallisto for pseudoalignment
 1. Downloaded from the following link on March 5, 2025:
 - a. https://github.com/pachterlab/kallisto-transcriptome-indices/releases/download/v1/mouse_index_standard.tar.xz
 - iii. sample_info_mouse_brain_rnaseq.xlsx (metadata, available in this folder, see above)
 - b. Outputs:
 - i. PCAs for each mouse brain RNA-seq experiment (there were two, see Methods) – used for a figure
 1. expt1_samples_pca.pdf
 2. expt2_samples_pca.pdf
 - ii. hemoglobin_genes_plot.pdf – used for a figure
 - iii. cooks_distance_tmao_samples.pdf – used for a figure
 - iv. DESeq2 results, used to create a supplementary table
 1. Contained within deseq2_outputs subfolder
 - v. Volcano plots for each metabolite tested (TMAO, PA, IAA, GDCA) – used for a figure
5. fgsea.Rmd
 - a. Inputs:
 - i. The DESeq2 results (produced by the previous script) for TMAO and GDCA
 - ii. The following collections of gene sets, downloaded on July 4, 2025 from: <https://www.gsea-msigdb.org/gsea/msigdb/mouse/collections.jsp>
 1. Hallmark gene sets – “mh.all.v2025.1.Mm.symbols.gmt”

2. REACTOME subset of CP (canonical pathways) –
“m2.cp.reactome.v2025.1.Mm.symbols.gmt”

b. Outputs:

- i. All contained within the fgsea_results subfolder:
 1. Results of gene set enrichment analysis, using the input collections of gene sets
 - a. Used for a supplementary table
 2. PDFs depicting selected significant terms from GSEA
 - a. Used for figures

6. marker_gene_analysis_tmao.Rmd

a. Inputs:

- i. Processed single-cell data from <https://doi.org/10.1038/s41593-019-0491-3>, downloaded from https://singlecell.broadinstitute.org/single_cell/study/SCP263/aging-mouse-brain
- ii. Genes significantly regulated by TMAO, as outputted by the DESeq2 script above

b. Outputs:

- i. Plot showing % of marker genes downregulated by TMAO, for each of six brain cell classes
 1. Used for a figure
- ii. RDS containing the computed marker genes per class
- iii. TSV containing the computed marker genes per class, used for a supplementary table
- iv. A TSV for a source data file

7. marker_gene_analysis_gdca.Rmd

a. Inputs:

- i. The RDS outputted by the previous script
- ii. Genes significantly regulated by GDCA, as outputted by the DESeq2 script above

b. Outputs:

- i. Plot showing % of marker genes downregulated by GDCA, for each of six brain cell classes
 1. Used for a figure
- ii. A TSV for a source data file

8. gaba_a_receptor_genes_heatmap.Rmd

a. Inputs:

- i. Same as those for pca_and_deseq2.Rmd, above

b. Outputs:

- i. A heatmap, used for a figure, showing downregulation of GABA_A receptor subunits in the brains of TMAO-treated mice
 - ii. The underlying numerical data for the heatmap
- 9. star_alignment_script.sh
 - a. Bash script that was used to run STAR alignment (instead of kallisto) on the trimmed fastq files, in order to prepare for differential splicing analysis with rMATS-turbo
- 10. star_index_script.sh
 - a. Bash script that was used to create the STAR index for STAR alignment
- 11. rmats_turbo_script.sh
 - a. Bash script that was used to run rMATS-turbo after STAR alignment
- 12. differential_splicing_events.Rmd
 - a. Inputs:
 - i. Files outputted by rMATS-turbo, available in this folder within the subfolder rmats_output
 - b. Outputs:
 - i. Excel files
 - 1. diff_splicing_analysis_results.xlsx
 - 2. sig_diff_splicing_events.xlsx (used for a supplementary table)
 - ii. Volcano plot used for a figure panel
 - 1. Along with a source data file
- 13. sashimi_code subfolder
 - a. Contains files used to generate supplementary sashimi plots of differential splicing in *Rsrp1*, *Cpsf4*, *Ythdf3*, and *Kmt2b* in brains from TMAO- vs. saline-treated mice.
 - b. Specifically contains:
 - i. Files containing the commands used to run ggsashimi.py (which was downloaded from <https://github.com/guigolab/ggsashimi> on June 26, 2025)
 - ii. Input files required for running those commands:
 - 1. palette_saline_grey.txt
 - 2. input_bams.tsv
 - iii. Raw output PDFs of the ggsashimi commands, which were later edited in Illustrator to reduce visual clutter and emphasize the junctions relevant to each differential splicing event
 - c. Required files not included in this repo:
 - i. Bam files (generated by STAR – very large)
 - ii. gencode.vM37.primary_assembly.sorted.gtf

1. This file was generated from gencode.vM37.primary_assembly.annotation.gtf, which was downloaded from GENCODE (release M37; GRCm39; <https://www.gencodegenes.org/mouse/>).
2. The annotation file was sorted on the command line using "sort -k1,1 -k4,4n"